Finite terms: BFKL terms and remaining terms

Contents. Sec. 1: `Evolution_terms.tex`, Sec. 1 + Sec. 2 vs. Sec. 3. Secs. 2–4: the terms of `Finite_terms_1.tex` that enter BFKL (its $\log \frac{\vee}{k^+}$ terms), row by row. Sec. 5: their sum. Secs. 6–9: the remaining terms, row by row, in the form of `Finite_terms_1.tex`. For every row: row = BFKL terms + remaining terms, exactly.

1 `Evolution_terms.tex`: does Sec. 1 + Sec. 2 give Sec. 3?

Yes, exactly, with no left-over. Every two-ρ term of the file was parsed and evaluated at random transverse points with random charges, with your convention $U^{bc} = U^{cb}$:

comparison	|difference|/|terms| (5 configurations)
Sec. 1 + Sec. 2 vs. Sec. 3, $d^3N_1 + d^3N_2$	7×10^{-17}
Sec. 1 + Sec. 2 vs. Sec. 3, collected form	8×10^{-17}
Sec. 3 vs. JIMWLK⊗LO	2×10^{-16}
new list of "From 4ρ's" (d^3N_2) vs. the two-ρ lists of `4rho.tex`	1×10^{-16}
Sec. 1 + Sec. 2 *without* the new list vs. Sec. 3	2.4%

- Sec. 1 has 139 terms (8 equations), Sec. 2 has 365 terms (16 equations). Their sum is Sec. 3 term by term, and Sec. 3 is JIMWLK⊗LO.
- The new list d^3N_2 of "From 4ρ's" was what was missing before: without it the difference is 2.4%.
- Two small points: in the colour-singlet form of Sec. 3, $\int_{x,,z,w,w'}$ should read $\int_{x,y,z,w,w'}$. For a general (non-symmetric) U only the collected form of Sec. 3 is exact; with $U^{bc} = U^{cb}$ all forms agree.
- Nothing in Secs. 1, 2 gives the BFKL bracket of Sec. 4: all $\log \frac{\vee}{\Lambda}$ terms add up to JIMWLK⊗LO alone. The BFKL bracket comes from `Finite_terms_1.tex`, with $\log \frac{\vee}{k^+}$ (Sec. 5).

2 BFKL terms: Group I (with c.c.)

2.1 Row I

NA

2.2 Row II

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\,2\times\log\Big[\frac{\vee}{k^+}\Big]\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times\right.$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^b(x) - U^{ab}(x)\,\rho^b(x)\Big] \tag{2.1}$$

$$\frac{d^3N_{2\,a}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2} - \frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]$$
$$+\,\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{abc}}{32\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}\right.$$
$$\left.-\,\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\Big] \tag{2.2}$$

Row II. Part 1: from $\log\big[\frac{\vee}{k^+}\big] + \log\big[\frac{\vee}{k^+} - 1\big] = 2\log\big[\frac{\vee}{k^+}\big] + \log\big[1 - \frac{k^+}{\vee}\big]$. Part 2a: all of it (nothing in it depends on z, so $\int_z$ is dropped). Part 2b: none.

2.3 Row III

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{$$

$$-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\} \tag{2.3}$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{$$

$$-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2}$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$+\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,y,z,x'}\Bigg[\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}$$

$$-\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2}$$

$$+\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}$$

$$-\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2}$$

$$+\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}$$

$$-\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\Bigg]$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\} \tag{2.4}$$

$$\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{$$

$$-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\vee}\Big]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\vee}\Big]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\left[(x-w)^2-(x-z)^2\right]^2}$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$+\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\vee}\Big]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \tag{2.5}$$

Row III. d^3N_1: from $\log\left|\frac{\vee(x-z)^2-k^+(x-w)^2}{(\Lambda+k^+)(x-z)^2-k^+(x-w)^2}\right|$, $\log\left[\frac{\vee}{k^+-\Lambda}\right]$ and $\log\left|\frac{k^+(x-w)^2-\vee(x-z)^2}{k^+(x-w)^2-\Lambda(x-z)^2}\right|$. The first two give $+\log\frac{\vee}{k^+}$ and $-\log\frac{\vee}{k^+}$ with the same kernel and cancel; the third gives the term above. d^3N_2: the first term from $\log\left[\frac{\vee}{\Lambda+k^+}\right]$, the six terms in the bracket from $\log\left|\frac{(x-w)^2k^+-(x-z)^2\vee}{(x-w)^2k^+-(x-z)^2(k^++\Lambda)}\right|$. d^3N_{3a}: from $\log\left[\frac{k^+-\Lambda}{\vee}\right] = -\log\frac{\vee}{k^+} + \log(1-\frac{\Lambda}{k^+})$. d^3N_{3b}: none.

2.4 Row IV

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]$$
$$+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}$$
$$\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big] \tag{2.6}$$

Row IV. The first term is from $\log\big[\frac{\vee}{\Lambda+k^+}\big]$, the second from

$$\log\Big[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}\Big]=\log\frac{\vee}{k^+}+\log\Big[\frac{(1-\frac{k^+}{\vee})(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{\frac{\Lambda}{k^+}(x-z)^2+(x-w)^2}\Big].$$

Together (the δ^{ij} part cancels the $\frac{1}{(z-w)^2}$ of the first term) the kernel is

$$\frac{ig^4}{8\pi^5}\frac{1}{k^+}\left[\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(z-w)^2(y-z)^2(x-z)^2}-\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(x-w)^2(y-z)^2(x-z)^2}-\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2(x-w)^2(y-z)^2(z-w)^2}\right],$$

which is integrable at $z\to w$; the first term alone is not, and with it alone the three-ρ terms could not cancel. The same holds for Group II Row I part 1 and Group III Rows I, II.

2.5 Row V

First part

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\Big]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\Big] \tag{2.7}$$

Second part

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.$$
$$\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y)$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}$$
$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y) \tag{2.8}$$

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y). \tag{2.9}$$

$$\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\vee}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.$$
$$\left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y)$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\vee}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.$$

$$+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\Big]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y)$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\vee}\right)\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times$$
$$\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y) \tag{2.10}$$

Row V. From $\log\left[\frac{k^+-\Lambda}{\vee}\right]=-\log\frac{\vee}{k^+}+\log(1-\frac{\Lambda}{k^+})$ and from the $-2\log\frac{\vee}{k^+}$ of 2b UV. The p^+, q^+ integrals (from 0 to k^+) have none.

3 BFKL terms: Group II

3.1 Row I

3.1.1 First part

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}$$
$$\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^{b}(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right]$$
$$-\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2}$$
$$\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^{c}(x'),\rho^{b}(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^{d}(x'),\rho^{e}(y)\}\right]$$
$$+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\left(-\frac{ig^4}{16\pi^5}\right)\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}$$
$$\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^{b}(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right]$$
$$+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}$$
$$\times\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]$$
$$\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^{c}(x'),\rho^{b}(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^{d}(x'),\rho^{e}(y)\}\right] \tag{3.1}$$

Row I, part 1. As Group I Row IV, in both blocks.

3.1.2 Second part

Row I, part 2. The logarithms $\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}\right]$ and $\log\left[\frac{(\vee-k^+)(x'-z)^2+k^+(x'-w')^2}{\Lambda(x'-z)^2+k^+(x'-w')^2}\right]$ each contain $\log\frac{\vee}{k^+}$. In 2b and in 2c the two terms of the row combine into the factors $\frac{1}{(x-w)^2(x'-w')^2}$ and $\frac{1}{(x-z)^2(x'-z)^2}$. 2a: none ($\vee$ cancels between numerator and denominator). 2d, 2e, 2f: their $\log\left[\frac{\vee}{k^++\Lambda}\right]$ cancels exactly against the $\log\frac{\vee}{k^+}$ of the two logarithms in the same rows: none.

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{(x-w)^2\,(x'-w')^2}$$
$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$- f^{ac'd'} f^{cbd} U^{ac}(w) U^{bc'}(z) U^{e'd'}(x') \rho^{e'}(x') \rho^{d}(x) + N_c U^{e'd}(x') U^{de}(x) \rho^{e'}(x') \rho^{e}(x) \Bigg] \tag{3.2}$$

$$\begin{aligned}
\frac{d^3 N_{2c\,\mathrm{NLO}}}{d^3k} = {} & \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^5} \frac{1}{k^+} \log\left[\frac{\vee}{k^+}\right] \int_{z,w,w',x,x'} e^{-ik(w'-w)} \frac{1}{(x-z)^2 (x'-z)^2} \\
& \left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} \right. \\
& \left. - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2} \right] \\
& \Big[f^{c'bd'} f^{cbd} U^{ac'}(w') U^{ac}(w) \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'} f^{acd} U^{ac'}(w') U^{bc}(z) U^{de}(x) \rho^{d'}(x') \rho^{e}(x) \\
& - f^{ac'd'} f^{cbd} U^{ac}(w) U^{bc'}(z) U^{e'd'}(x') \rho^{e'}(x') \rho^{d}(x) + N_c U^{e'd}(x') U^{de}(x) \rho^{e'}(x') \rho^{e}(x) \Big]
\end{aligned} \tag{3.3}$$

The first term of d^3N_{2c} has the kernel $\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$: the measured gluon is emitted by the soft gluon z in the amplitude and in the conjugate amplitude (second line of your Eq. (LOBFKL)).

3.2 Row II

NA

3.3 Row III

NA

3.4 Row IV

$$\begin{aligned}
\frac{d^3 N_{\mathrm{NLO}}}{d^3k} = {} & - \frac{g^2}{4\pi^2} \frac{N_c}{(2\pi)^3} \frac{g^2}{2\pi^2} \frac{1}{k^+} \log\left[\frac{\vee}{k^+}\right] \int_{x,y,x',y'} e^{-ik\cdot(y'-y)} \frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2} \\
& \left[-\frac{d_\perp}{\epsilon} - d_\perp\,\gamma_E - \frac{d_\perp}{2} \log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4} \right) - 1 \right] \Big[U^{e'c}(y') - U^{e'c}(x') \Big] \times \left[U^{ec}(y) - U^{ce}(x) \right] \rho^{e'}(x') \rho^{e}(x) \\
& + \frac{1}{(2\pi)^3} \frac{g^4\,N_c}{8\pi^5} \frac{1}{k^+} \log\left[\frac{\vee}{k^+}\right] \int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)} \frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2} \frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2} \\
& \Big[U^{e'c}(z) - U^{e'c}(x') \Big] \times \left[U^{ec}(z) - U^{ce}(x) \right] \rho^{e'}(x') \rho^{e}(x)
\end{aligned} \tag{3.4}$$

Row IV. Both terms come from the lower end $\xi \to \frac{k^+}{\vee}$ of $\int_{k^+/\vee}^1 d\xi$ in `Finite_terms_1.tex`:

- first term: your subtraction is $\frac{1}{\xi^2}\frac{\xi}{(1-\xi)_+} e^{-ik\cdot(y'-y)\frac{1}{\xi}} = \frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}} - e^{-ik\cdot(y'-y)}}{\xi(1-\xi)}$. With $\frac{1}{\xi(1-\xi)} = \frac{1}{\xi} + \frac{1}{1-\xi}$, the part $-\frac{e^{-ik\cdot(y'-y)}}{\xi}$ gives $-e^{-ik\cdot(y'-y)} \int_{k^+/\vee}^1 \frac{d\xi}{\xi} = -e^{-ik\cdot(y'-y)} \log\frac{\vee}{k^+}$; the rest has no $\log\frac{\vee}{k^+}$. Together with Group I Row II part 1 (coefficient 2) and Row V 2b UV, its $\frac{1}{\epsilon}$ cancels: no pole is left in this list.
- second term: the term $\frac{1-\xi}{\xi}$ of P_{gg} at small ξ. There $e^{-ik\cdot(y'-y)/\xi}$ probes momentum $\frac{k}{\xi}$, and $\int d^2(y'-y)\, e^{-iq\cdot(y'-y)} \log(y'-y)^2\, h(y'-y) \to -\frac{4\pi}{q^2} h(0)$ puts $y = y' = z$ and gives a factor ξ^2, so $\int_{k^+/\vee} \frac{d\xi}{\xi} = \log\frac{\vee}{k^+}$. The factor $\frac{4\pi^2}{k^2}$ is written as $\int_{w,w'} e^{-ik\cdot(w'-w)} \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$. This term is BFKL: the measured gluon is emitted by the gluon at z on both sides.

4 BFKL terms: Group III (with c.c.)

4.1 Row I

$$\begin{aligned}
\frac{d^3 N_{\mathrm{NLO}}}{d^3k} = {} & \frac{1}{(2\pi)^3} \int_{x,y,x',z,w,w'} e^{-ik(w'-w)} \Bigg\{ \\
& \frac{ig^4}{16\pi^5} \frac{1}{k^+} \log\left[\frac{\vee}{k^+}\right] \frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \\
& \Big[U^{ab'}(x')\, U^{c'd'}(y)\, U^{c'b}(z)\, \rho^{b'}(x') \rho^{d'}(y) - U^{bd'}(z)\, U^{d'e'}(y) U^{ac'}(w')\, \rho^{c'}(x') \rho^{e'}(y) \Big] \times
\end{aligned}$$

$$
\Big[f^{cbd}\, U^{ac}(w)\, \rho^d(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^e(x) \Big]
$$

$$
+ \frac{1}{(2\pi)^3} \int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)} \Big(- \frac{ig^4}{8\pi^5} \Big) \frac{1}{k^+} \log\Big[\frac{\vee}{k^+} \Big] \frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}
$$

$$
\times \left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} \right]
$$

$$
\Big[U^{ab'}(x')\, U^{c'd'}(y)\, U^{c'b}(z)\, \rho^{b'}(x') \rho^{d'}(y) - U^{bd'}(z)\, U^{d'e'}(y) U^{ac'}(w')\, \rho^{c'}(x') \rho^{e'}(y) \Big] \times
$$

$$
\Big[f^{cbd}\, U^{ac}(w)\, \rho^d(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^e(x) \Big] \tag{4.1}
$$

4.2 Row II

$$
\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\, \frac{ig^4}{16\pi^5}\, \frac{1}{k^+}\, \log\Big[\frac{\vee}{k^+} \Big]\, \frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \times
$$

$$
\Big[U^{c'd'}(y)\, U^{ab'}(x')\, U^{bc'}(z)\, \rho^{d'}(y) \rho^{b'}(x') - U^{ab'}(x')\, \rho^b(y) \rho^{b'}(x') \Big] \times \Big[f^{cbd}\, U^{ac}(w)\, \rho^d(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^e(x) \Big]
$$

$$
+ \frac{1}{(2\pi)^3} \int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)} \Big(- \frac{ig^4}{8\pi^5} \Big) \frac{1}{k^+} \log\Big[\frac{\vee}{k^+} \Big] \frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}
$$

$$
\times \left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} \right]
$$

$$
\Big[U^{c'd'}(y)\, U^{ab'}(x')\, U^{bc'}(z)\, \rho^{d'}(y) \rho^{b'}(x') - U^{ab'}(x')\, \rho^b(y) \rho^{b'}(x') \Big] \times
$$

$$
\Big[f^{cbd}\, U^{ac}(w)\, \rho^d(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^e(x) \Big] \tag{4.2}
$$

Rows I, II. The first term is from $\log\big[\frac{\vee}{k^+ + \Lambda}\big]$, the second from the $\log \frac{\vee}{k^+}$ inside $\log\Big[\frac{(\vee - k^+)(x-z)^2 + k^+ (x-w)^2}{k^+ (x-w)^2 + \Lambda (x-z)^2}\Big]$, as in Group I Row IV.

4.3 Row III

4.3.1 First part

$$
\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3}\, \frac{ig^4\, f^{c'd'a}}{16\pi^5}\, \frac{1}{k^+}\, \log\Big[\frac{\vee}{k^+} \Big] \int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\, \frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\Big[U^{e'c'}(y')\, U^{d'b}(z)\, \rho^{e'}(x') - U^{bd'}(z)\, U^{c'e'}(x')\, \rho^{e'}(x') \Big] \times
$$

$$
\big[U^{ac}(w)\, \{\rho^c(y), \rho^b(x)\} - U^{bc}(z)\, U^{ad}(y)\, U^{ce}(x)\, \{\rho^d(y), \rho^e(x)\} \big] \tag{4.3}
$$

4.3.2 Second part

$$
\frac{d^3 N_{2\,\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+}\, \log\Big[\frac{\vee}{k^+} \Big] \int_{w,x',y',z,x} e^{-ik\cdot(y'-w)} \left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2} \right.
$$

$$
\left. + \frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2} \right] \Big[f^{c'd'a}\, U^{e'c'}(y')\, U^{d'b}(z)\, \rho^{e'}(x') - f^{c'd'a}\, U^{bd'}(z)\, U^{c'e'}(x')\, \rho^{e'}(x') \Big] \times
$$

$$
\big[f^{cbd}\, U^{ac}(w)\, \rho^d(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^e(x) \big] \tag{4.4}
$$

4.4 Row IV

NA

4.5 Row V

$$
\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3}\, \frac{ig^4\, f^{c'ca}}{8\pi^5}\, \frac{1}{k^+}\, \log\Big[\frac{\vee}{k^+} \Big] \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\, \frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times$$
$$\big[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\big] \tag{4.5}$$

4.6 Row VI

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times$$
$$\big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\big] \tag{4.6}$$

Group III Rows III, V, VI. From the $\Big[\frac{1}{p^+}\Big]_+\delta_{jm}\delta_{ik'}$ term:

$$\int_0^{\vee-k^+} dp^+\,\Big[\frac{1}{p^+}\Big]_+ e^{-ik\cdot(y'-z)\frac{p^+}{k^+}} = -\Big[\log\frac{\vee}{k^+} + \log\Big(1-\frac{k^+}{\vee}\Big) + \gamma_E + \log\big(ik\cdot(y'-z)\big) + E_1\Big(ik\cdot(y'-z)\frac{\vee-k^+}{k^+}\Big)\Big]$$

(in part 2 in the same way, with $\frac{1}{p^+(x-z)^2+k^+(x-w)^2} \to \frac{1}{k^+(x-w)^2}$ at $p^+=0$). The other terms of these rows have no $\log\frac{\vee}{k^+}$.

Why the $\Big[\frac{1}{p^+}\Big]_+$ term contains $\log\frac{\vee}{k^+}$. $\Big[\frac{1}{p^+}\Big]_+ f = \frac{f(p^+)-f(0)}{p^+}$ is finite at $p^+\to 0$: this term has no $\log\Lambda$. But the p^+ integral runs up to $\vee-k^+$, and the subtraction $-\frac{f(0)}{p^+}$ falls only like $\frac{1}{p^+}$. Splitting at $p^+=k^+$, with $f(p^+)=e^{-ik\cdot(y'-z)p^+/k^+}$:

$$\int_0^{\vee-k^+} dp^+\,\frac{f(p^+)-f(0)}{p^+} = \underbrace{\int_0^{k^+} dp^+\,\frac{f(p^+)-f(0)}{p^+} + \int_{k^+}^{\vee-k^+} dp^+\,\frac{f(p^+)}{p^+}}_{\text{finite for }\vee\to\infty\ (f\text{ oscillates})} - f(0)\log\Big(\frac{\vee}{k^+}-1\Big).$$

The p^+ integral done numerically (no closed form used; `pplus_check.py`), $k\cdot(y'-z)=0.7$:

$\frac{\vee}{k^+}$	$I=\int_0^{\vee-k^+}dp^+\,\Big[\frac{1}{p^+}\Big]_+ f$	$I+\log\frac{\vee}{k^+}$
10^1	$-2.4377-1.4182\,i$	$-0.13507-1.41817\,i$
10^2	$-4.8132-1.5566\,i$	$-0.20804-1.55658\,i$
10^3	$-7.1259-1.5712\,i$	$-0.21817-1.57121\,i$
10^4	$-9.4308-1.5707\,i$	$-0.22046-1.57066\,i$
10^5	$-11.7335-1.5708\,i$	$-0.22054-1.57080\,i$
10^6	$-14.0360-1.5708\,i$	$-0.22054-1.57080\,i$
∞		$-\gamma_E-\log\big(ik\cdot(y'-z)\big) = -0.22054-1.57080\,i$

I grows like $-\log\frac{\vee}{k^+}$; $I+\log\frac{\vee}{k^+}$ is finite. The same holds in part 2 (`pplus_check.json`). This $\log\frac{\vee}{k^+}$ belongs where it is put: before the subtraction the same term was $f(0)\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+} + \int_0^{\vee-k^+}dp^+\,\Big[\frac{1}{p^+}\Big]_+ f$. Its $f(0)\log\frac{\vee}{\Lambda}$ is in `Evolution_terms.tex`, its $f(0)\log(1-\frac{k^+}{\vee})$ is the $\log(1-\frac{k^+}{\vee})$ line of the row, and all three together give

$$f(0)\Big[\log\frac{\vee}{\Lambda} + \log\Big(1-\frac{k^+}{\vee}\Big) - \log\Big(\frac{\vee}{k^+}-1\Big)\Big] + \text{finite} = f(0)\log\frac{k^+}{\Lambda} + \text{finite}.$$

The phase $e^{-ik\cdot(y'-z)p^+/k^+}$ cuts the p^+ integral off at $p^+\sim k^+/|k\cdot(y'-z)|$, so this term really has $\log\frac{k^+}{\Lambda}$, not $\log\frac{\vee}{\Lambda}$; the $\log\frac{\vee}{k^+}$ of the $\Big[\frac{1}{p^+}\Big]_+$ term removes the difference. By contrast, Group I Row V has $\Big[\frac{k^+}{p^+}\Big]_+$ on $0<p^+<k^+$: no $\vee$, so nothing is taken out there.

5 Sum of the BFKL terms

With $\mathcal{N}\equiv\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$, $\mathcal{J}$ = the bracket of Sec. 3 of `Evolution_terms.tex` (JIMWLK⊗LO) and $\mathcal{B}$ = the bracket of its Sec. 4 (BFKL⊗LO, your Eq. (LOBFKL)):

1. The three-ρ terms of Secs. 2–4 cancel after symmetrization (checked: `LogVeeKplus_BFKL.pdf`, all 39 structures).
2. The two-ρ terms of Secs. 2–4 add up to
$$\sum_{\text{Secs. 2–4}}\frac{d^3N}{d^3k} = \mathcal{N}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\mathcal{B}-\mathcal{J}\Big]. \tag{5.1}$$
3. The reason for $-\mathcal{J}$: every row has the soft gluon integrated over $\Lambda<p^+<\vee$. With T the integrand for $p^+\ll k^+$ and P the integrand for $p^+\gg k^+$,
$$\int_\Lambda^\vee\frac{dp^+}{p^+}\,G\Big(\frac{p^+}{k^+}\Big) = T\log\frac{k^+}{\Lambda} + P\log\frac{\vee}{k^+} + \dots = T\log\frac{\vee}{\Lambda} + (P-T)\log\frac{\vee}{k^+} + \dots$$
$T\log\frac{\vee}{\Lambda}$ is in `Evolution_terms.tex`, and $\sum T=\mathcal{J}$ (Sec. 1 + Sec. 2 = Sec. 3, Sec. 1 above); $\sum P=\mathcal{B}$. So the $\log\frac{\vee}{k^+}$ terms are $\sum(P-T)=\mathcal{B}-\mathcal{J}$.

4. Both files together:

$$\boxed{\mathcal{N}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\log\Big[\frac{\vee}{\Lambda}\Big]\mathcal{J}+\log\Big[\frac{\vee}{k^+}\Big](\mathcal{B}-\mathcal{J})\Big]=\mathcal{N}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\log\Big[\frac{k^+}{\Lambda}\Big]\mathcal{J}+\log\Big[\frac{\vee}{k^+}\Big]\mathcal{B}\Big].} \tag{5.2}$$

JIMWLK comes with $\log\frac{k^+}{\Lambda}$ and BFKL with $\log\frac{\vee}{k^+}$. In Sec. 4 of `Evolution_terms.tex` the bracket is right, but its logarithm is $\log\frac{\vee}{k^+}$; in Sec. 3 the logarithm becomes $\log\frac{k^+}{\Lambda}$ once the $-\mathcal{J}\log\frac{\vee}{k^+}$ above is added to it. At mid rapidity, $k^+=\sqrt{\vee\Lambda}$, both logarithms are $\frac{1}{2}\log\frac{\vee}{\Lambda}$.

6 Remaining terms: how they are obtained

Every row of `Finite_terms_1.tex` is, exactly (any $\vee$, any Λ),

$$\text{row} = \big(\text{its BFKL terms, Secs. 2–4}\big) + \big(\text{its remaining terms, Secs. 7–9}\big).$$

The remaining terms below are the rows of `Finite_terms_1.tex` in the same form, with these changes (blue: changed or new; red: removed):

- every logarithm that contains $\vee$ is replaced by what is left of it after $\log\frac{\vee}{k^+}$ is taken out ($A=(x-z)^2$, $B=(x-w)^2$, and the same with x', w'):

$$\log\Big[\frac{(\vee-k^+)A+k^+B}{\Lambda A+k^+B}\Big]=\log\frac{\vee}{k^+}+\log\Big[\frac{\big(1-\frac{k^+}{\vee}\big)A+\frac{k^+}{\vee}B}{\frac{\Lambda}{k^+}A+B}\Big],$$

$$\log\Big[\frac{\vee}{k^++\Lambda}\Big]=\log\frac{\vee}{k^+}-\log\Big[1+\frac{\Lambda}{k^+}\Big],\qquad \log\Big[\frac{\vee}{k^+-\Lambda}\Big]=\log\frac{\vee}{k^+}-\log\Big[1-\frac{\Lambda}{k^+}\Big],$$

$$\log\Big[\frac{k^+-\Lambda}{\vee}\Big]=-\log\frac{\vee}{k^+}+\log\Big[1-\frac{\Lambda}{k^+}\Big],\qquad \log\Big[\frac{\vee}{k^+}-1\Big]=\log\frac{\vee}{k^+}+\log\Big[1-\frac{k^+}{\vee}\Big],$$

$$\log\Big|\frac{\vee A-k^+B}{(\Lambda+k^+)A-k^+B}\Big|=\log\frac{\vee}{k^+}+\log\Big|\frac{A-\frac{k^+}{\vee}B}{\big(1+\frac{\Lambda}{k^+}\big)A-B}\Big|,$$

$$\log\Big|\frac{k^+B-\vee A}{k^+B-\Lambda A}\Big|=\log\frac{\vee}{k^+}+\log\Big|\frac{\frac{k^+}{\vee}B-A}{B-\frac{\Lambda}{k^+}A}\Big|,$$

$$\log\Big|\frac{Bk^+-A\vee}{Bk^+-A(k^++\Lambda)}\Big|=\log\frac{\vee}{k^+}+\log\Big|\frac{B\frac{k^+}{\vee}-A}{B-A\big(1+\frac{\Lambda}{k^+}\big)}\Big|;$$

- Group II Row IV and Group III Rows III, V, VI: the ξ and p^+ integrals that contain $\log\frac{\vee}{k^+}$ are done (notes under these rows).

All identities are checked numerically to 10^{-14} (`split_logs.json`). In the remaining terms $\vee$ enters only through $\frac{k^+}{\vee}$, $\log(1-\frac{k^+}{\vee})$, $\int_{k^+/\vee}d\xi$, $\frac{1}{\vee}$ and the bounded oscillating functions $E_1\big(ik\cdot(y'-z)\frac{\vee-k^+}{k^+}\big)$, $\int_0^{\vee-k^+}dp^+$ of Group III: they have a finite limit for $\vee\to\infty$. The one exception is written out: $\log(1-\frac{k^+}{\vee})\log\frac{\vee}{k^+}$ in Group I Row II, which vanishes like $\frac{k^+}{\vee}\log\frac{\vee}{k^+}$. $\Lambda\to 0$ can be taken in all of them.

7 Remaining terms: Group I (with c.c.)

7.1 Row I

NA

7.2 Row II

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x',}e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\Bigg[\Bigg[-\frac{2}{\epsilon}-2\gamma-\log\Bigg[\frac{(x-w)^2\,\mu_{\overline{MS}}^2}{4}\Bigg]\Bigg]\times$$
$$\Big[\frac{11}{3}+\log\Big[1-\frac{k^+}{\vee}\Big]\Big]-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\Big[1-\frac{k^+}{\vee}\Big]+\log\Big[1-\frac{k^+}{\vee}\Big]\log\Big[\frac{\vee}{k^+}\Big]\Bigg]\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\big] \tag{7.1}$$

d^3N_{2a}: no remaining term (it is proportional to $\log\frac{\vee}{k^+}$; all of it is in Sec. 2).

$$\frac{d^3N_{2\,b}}{d^3k}=-\frac{1}{(2\pi)^3}\frac{g^4}{64\pi^6}\frac{1}{k^+}\int_{w,w'}e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y}e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})$$

$$\times \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\} - f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\Big] \tag{7.2}$$

where

$$R^i = -2k^i \log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}\,p^2}\right) + \frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right] - \frac{p^2k^i+k^2p^i}{(k-p)^2}\,\log\left(1-\frac{k^+}{\vee}\right). \tag{7.3}$$

Part 1: the remaining $\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]$ vanishes for $\vee\to\infty$; it is not a $\log\frac{\vee}{k^+}$ term. Part 2b: no change.

7.3 Row III

$$\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2\right]}{\left[\left(1+\frac{\Lambda}{k^+}\right)(x-z)^2-(x-w)^2\right]}\right] \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left(-\log\left[1-\frac{\Lambda}{k^+}\right]\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[\Lambda\,(x-z)^2-k^+\,(x-w)^2\right]}\right] \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{\left[\frac{k^+}{\vee}\,(x-w)^2-(x-z)^2\right]}{\left[(x-w)^2-\frac{\Lambda}{k^+}\,(x-z)^2\right]}\right] \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
\end{aligned} \tag{7.4}$$

$$\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
\end{aligned}$$

$$
\begin{aligned}
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{7.5}
$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Bigg\{
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2}
\end{aligned}
$$

$$
\begin{aligned}
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'} \frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \Bigg\}
\end{aligned}
\tag{7.6}
$$

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'} \frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w) - (k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w) - \Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w) - (k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w) - \Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w) - (k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w) - \Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \Bigg\}
\end{aligned}
\tag{7.7}
$$

where

$$
\begin{aligned}
\mathcal{A} = \frac{1}{\sqrt{(x-w)^2(x-z)^2 - \left[(x-w)\cdot(x-z)\right]^2}} &\Bigg\{ \arctan\left[\frac{(k^+-\Lambda)(x-z)^2 - k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2 - \left[(x-w)\cdot(x-z)\right]^2}}\right] \\
&- \arctan\left[\frac{\Lambda(x-z)^2 - k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2 - \left[(x-w)\cdot(x-z)\right]^2}}\right]\Bigg\}
\end{aligned}
$$

7.4 Row IV

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)} \Bigg\{ \\
&\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}(x-w)^2}{\frac{\Lambda}{k^+}(x-z)^2 + (x-w)^2}\right] \times \left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
&\left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right] \\
&- \frac{ig^4}{16\pi^5}\frac{1}{k^+}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \times \left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right) \\
&+ \frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[1-\frac{k^+}{\vee}\right]\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]
\end{aligned}
\tag{7.8}
$$

7.5 Row V

First part

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\left[\delta_{im}\,\delta_{jk}-\left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\\
&\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\left[\frac{1}{q^+}\right]_+\\
&\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]\\
&-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]\\
&-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]
\end{aligned}
\tag{7.9}
$$

Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.\\
&\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}
\end{aligned}
$$

$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{7.10}$$

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}\right]\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y). \tag{7.11}$$

$$\begin{aligned}
\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k}&=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^\Lambda\frac{dq^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'}e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.\\
&\left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}+2\log\left[1-\frac{\Lambda}{k^+}\right]\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned} \tag{7.12}$$

8 Remaining terms: Group II

8.1 Row I

8.1.1 First part

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{$$

$$
- \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times
$$

$$
\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]
$$

$$
+\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}
$$

$$
-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}
$$

$$
\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
$$

$$
+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{
$$

$$
\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w')^2}\right]\times
$$

$$
\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]
$$

$$
-\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2}
$$

$$
+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}
$$

$$
\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] \tag{8.1}
$$

8.1.2 Second part

$$
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\Bigg\{\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times
$$

$$
\left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right.
$$

$$
+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}
$$

$$
+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}
$$

$$
\left.-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\right]\times
$$

$$
\left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{\Lambda\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x'-w')^2+\Lambda\,(x'-z)^2}\right]\right]\Bigg\}\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right] \tag{8.2}
$$

$$
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times
$$

$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$

$$
\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times
$$

$$
\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big]
$$

$$
- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2 - (x-w)^4\,(x'-z)^2]} \times
$$

$$
\Big[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} + \frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}
$$

$$
+ \frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2} \Big]
$$

$$
\times \log \Big[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2 + (x-w)^2} \Big] \times
$$

$$
\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big]
$$

$$
+ \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\, - (x-w)^2\,(x'-z)^2\,(x'-w')^2]} \times
$$

$$
\Big[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} + \frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}
$$

$$
+ \frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2} \Big]
$$

$$
\times \log \Big[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2 + \frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2 + (x'-w')^2} \Big]
$$

$$
\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big] \tag{8.3}
$$

$$
\frac{d^3 N_{2c\,\mathrm{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2}{(x'-z)^2}\, \frac{1}{[(x'-w')^2\,(x-z)^2 - (x-w)^2\,(x'-z)^2]}
$$

$$
\Big[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}
$$

$$
+ \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2} \Big] \times \log \Big[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2 + \frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2 + (x'-w')^2} \Big]
$$

$$
\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big]
$$

$$
- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-w)^2}{(x-z)^2}\, \frac{1}{[(x'-w')^2\,(x-z)^2 - (x-w)^2\,(x'-z)^2]}
$$

$$
\Big[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}
$$

$$
+ \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2} \Big] \times \log \Big[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2 + (x-w)^2} \Big]
$$

$$\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$

$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg] \tag{8.4}$$

$$\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k}=&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}\\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2+(x'-w')^2}\right]\times\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
\end{aligned} \tag{8.5}$$

$$\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}\\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
\end{aligned}$$

$$
\begin{aligned}
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-w)^2}{\left[(x-z)^2-(x-w)^2\right]}\, \frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2} \\
&+ \frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times \\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2}{\left[(x'-w')^2-(x'-z)^2\right]}\, \frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2} \\
&+ \frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times \\
&\log\left[\frac{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}{(x'-w')^2+\frac{\Lambda}{k^+}\,(x'-z)^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big]
\end{aligned}
\tag{8.6}
$$

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5} \left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right) \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\left[(x'-z)^2-(x'-w')\right.} \\
&\Big[f^{c'bd'}\, U^{ac'}(w')\, \rho^{d'}(x') - f^{ac'd'}\, U^{bc'}(z) U^{e'd'}(x')\, \rho^{e'}(x')\Big] \times \Big[f^{cbd}\, U^{ac}(w)\, \rho^{d}(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\, \frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5} \left[\frac{1}{\vee} - \frac{1}{k^+ + \Lambda}\right] \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{1}{(z-w)^2\,(z-w')^2} \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}
\end{aligned}
$$

$$
\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\left[(x'-w')^2(x-z)^2-(x-w)^2(x'-z)^2\right]}
$$
$$
\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{\frac{\Lambda}{k^+}(x'-z)^2+(x'-w')^2}\right]\times
$$
$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$
$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{8.7}
$$

8.2 Row II

NA

8.3 Row III

NA

8.4 Row IV

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]
$$
$$
\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\right]+\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{\frac{1}{\xi}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{1-\xi}\right]
$$
$$
\times\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)
$$
$$
+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.
$$
$$
\left.-\frac{d_\perp}{2}\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)
$$
$$
+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\,d_\perp\right]
$$
$$
\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)
$$
$$
-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}
$$
$$
\Big[U^{e'c}(z)-U^{e'c}(x')\Big]\times\left[U^{ec}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x) \tag{8.8}
$$

Row IV. Your subtraction $\frac{1}{\xi^2}\frac{\xi}{(1-\xi)_+}e^{-ik\cdot(y'-y)\frac{1}{\xi}}=\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{\xi(1-\xi)}$ (red) is written as $\frac{\frac{1}{\xi}e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{1-\xi}-\frac{e^{-ik\cdot(y'-y)}}{\xi}$; the first part is the blue ξ integral, the second gives $-e^{-ik\cdot(y'-y)}\log\frac{\vee}{k^+}$, which is in the BFKL terms. The first two lines are now the standard $\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,{\rm LO}_\xi$. At its lower end $\xi\to\frac{k^+}{\vee}$ the ξ integral still produces $\log\frac{\vee}{k^+}$ times the integral of the last two (blue) lines, from the $\frac{1-\xi}{\xi}$ term at small ξ; this is the second BFKL term of this row (Sec. 3). The last two lines subtract it, so the sum is finite for $\vee\to\infty$.

9 Remaining terms: Group III (with c.c.)

9.1 Row I

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\Bigg\{
$$
$$
-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2+\frac{\Lambda}{k^+}(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right.
$$
$$
\left.-\frac{(x-z)^j(x-w)^i}{(x-w)^2(x-z)^2}-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}\right]
$$

$$
\begin{aligned}
&+\frac{ig^4}{16\pi^5}\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\\
&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{9.1}
$$

9.2 Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\Bigg\{
&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2+\frac{\Lambda}{k^+}(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.\\
&\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]\\
&+\frac{ig^4}{16\pi^5}\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\\
&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{9.2}
$$

9.3 Row III

9.3.1 First part

$$
\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]\\
&-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\log\left[1-\frac{k^+}{\vee}\right]\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]\\
&+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\left[\gamma_E+\log\left(ik\cdot(y'-z)\right)+E_1\left(ik\cdot(y'-z)\,\frac{\vee-k^+}{k^+}\right)\right]\\
&\times\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]
\end{aligned}
\tag{9.3}
$$

The $\left[\frac{1}{p^+}\right]_+$ term (red) is integrated: $-\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\left[\frac{1}{p^+}\right]_+e^{-ik\cdot(y'-z)\frac{p^+}{k^+}}=\frac{1}{2\pi}\Big[\log\frac{\vee}{k^+}+\log\left(1-\frac{k^+}{\vee}\right)+\gamma_E+\log\left(ik\cdot(y'-z)\right)+E_1\big(ik\cdot(y'-z)\frac{\vee-k^+}{k^+}\big)\Big]$. Its $\log\frac{\vee}{k^+}$ is in the BFKL terms; its $\log(1-\frac{k^+}{\vee})$ cancels the red last line; the rest is the blue term.

9.3.2 Second part

$$
\frac{d^3N_{2\,\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}
$$

$$
\begin{aligned}
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.\\
&\left.\left.-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]\right.\\
&-\left[\frac{1}{p^+}\right]_+\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\\
&\left.-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,x}e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.\\
&\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,x',y',z,x}e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\\
&\times\left[\gamma_E+\log\left(ik\cdot(y'-z)\right)+E_1\Big(ik\cdot(y'-z)\,\frac{\vee-k^+}{k^+}\Big)\right.\\
&\quad\left.+e^{ik\cdot(y'-z)\frac{(x-w)^2}{(x-z)^2}}\left[E_1\Big(ik\cdot(y'-z)\,\frac{(x-w)^2}{(x-z)^2}\Big)-E_1\Big(ik\cdot(y'-z)\Big[\frac{(x-w)^2}{(x-z)^2}+\frac{\vee-k^+}{k^+}\Big]\Big)\right]\right]\\
&\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{9.4}
$$

The $\left[\frac{1}{p^+}\right]_+$ term (red) is integrated, with $f(p^+)=\frac{e^{-ik\cdot(y'-z)p^+/k^+}}{p^+(x-z)^2+k^+(x-w)^2}$: $\int_0^{\vee-k^+}dp^+\frac{f(p^+)-f(0)}{p^+}=-\frac{1}{k^+(x-w)^2}\Big[\log\frac{\vee}{k^+}+\log\left(1-\frac{k^+}{\vee}\right)+\gamma_E+\log\left(ik\cdot(y'-z)\right)+E_1\big(ik\cdot(y'-z)\frac{\vee-k^+}{k^+}\big)+\mathcal{J}\Big]$, $\mathcal{J}=$ the last bracket of the blue term. Its $\log\frac{\vee}{k^+}$ is in the BFKL terms; its $\log(1-\frac{k^+}{\vee})$ cancels the red last line.

9.4 Row IV

NA

9.5 Row V

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\left[\gamma_E+\log\left(ik\cdot(y'-z)\right)+E_1\Big(ik\cdot(y'-z)\,\frac{\vee-k^+}{k^+}\Big)\right]\\
&\times\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]
\end{aligned}
\tag{9.5}
$$

The $\left[\frac{1}{p^+}\right]_+$ term (red) is integrated: $-\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\left[\frac{1}{p^+}\right]_+e^{-ik\cdot(y'-z)\frac{p^+}{k^+}}=\frac{1}{2\pi}\Big[\log\frac{\vee}{k^+}+\log\left(1-\frac{k^+}{\vee}\right)+\gamma_E+\log\left(ik\cdot(y'-z)\right)+E_1\big(ik\cdot(y'-z)\frac{\vee-k^+}{k^+}\big)\Big]$. Its $\log\frac{\vee}{k^+}$ is in the BFKL terms; its $\log(1-\frac{k^+}{\vee})$ cancels the red last line; the rest is the blue term.

9.6 Row VI

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\left[\gamma_E + \log\big(ik\cdot(y'-z)\big) + E_1\Big(ik\cdot(y'-z)\,\frac{\vee-k^+}{k^+}\Big)\right]\\
&\times\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\right]
\end{aligned}
\tag{9.6}
$$

The $\left[\frac{1}{p^+}\right]_+$ term (red) is integrated: $-\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\left[\frac{1}{p^+}\right]_+ e^{-ik\cdot(y'-z)\frac{p^+}{k^+}} = \frac{1}{2\pi}\Big[\log\frac{\vee}{k^+} + \log\big(1-\frac{k^+}{\vee}\big) + \gamma_E + \log\big(ik\cdot(y'-z)\big) + E_1\big(ik\cdot(y'-z)\frac{\vee-k^+}{k^+}\big)\Big]$. Its $\log\frac{\vee}{k^+}$ is in the BFKL terms; its $\log(1-\frac{k^+}{\vee})$ cancels the red last line; the rest is the blue term.